Nowcasting

Answers

Workshop 2 – Workshop on Useful Statistical Procedure for Nowcasting Part 2: Stationarity



IMF Institute for Capacity Development

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The dataset for this workshop is in "**stationarity.wf1**". This file contains three time series of daily (5-day per week) exchange rates: the exchange values of the Canadian dollar (**cadusd**), the British pound (**gbpusd**), and the Swiss franc (**chfusd)** against the US dollar, from January 4, 1971 to February 20, 2015 (44 years). In what follows, use only the data for the period January 1, 1976 to December 31, 2009 (you have additional observations in the workfile).

Question 1. Plot the ***(logarithm of)*** each of the resulting series on a single graph and "eyeball" them to decide whether they might be stationary or not.

Generate **lcadusd = log(cadusd)** by double clicking **cadusd**, choosing Proc – Generate by Equation, typing the equation and clicking OK. Repeat for the remaining two exchange rate series. We can also generate the series using the following commands:

```
Smpl @all

genr lchfusd = log(chfusd)

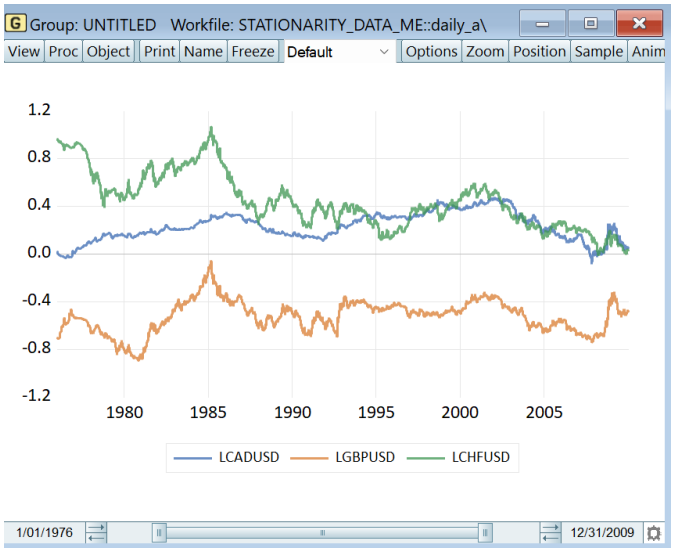
genr lgbpusd = log(gbpusd)

genr lcadusd = log(cadusd)
```

After the three series are all generated, select the three series by holding the Ctrl key down and clicking on each series. Then right click on the selected series, choose "Open -> as Group", and then View->Graph. To change sample, double click "Sample:" in the work file window and type in the required sample in the box (1/1/1976 12/31/2009).

Or simply use the type in the following code and hit enter after each command:

```
smpl 1/1/1976 12/31/2009
plot lcadusd lchfusd lgbpusd
```



All three exchange rate series show a tendency to drift over time without returning to a consistent mean or trend. This behavior suggests the presence of *stochastic trends*, a common feature of non-stationary time series. The series do not fluctuate randomly around a fixed mean or deterministic trend. Instead, they appear to *evolve in a random walk-like manner*, which is indicative of non-stationarity.

For the rest of this workshop, we shall ***focus only on the logarithm of the Swiss Franc-US dollar series, LCHFUSD***. (You can perform the same tasks below for the other series later if you want.)

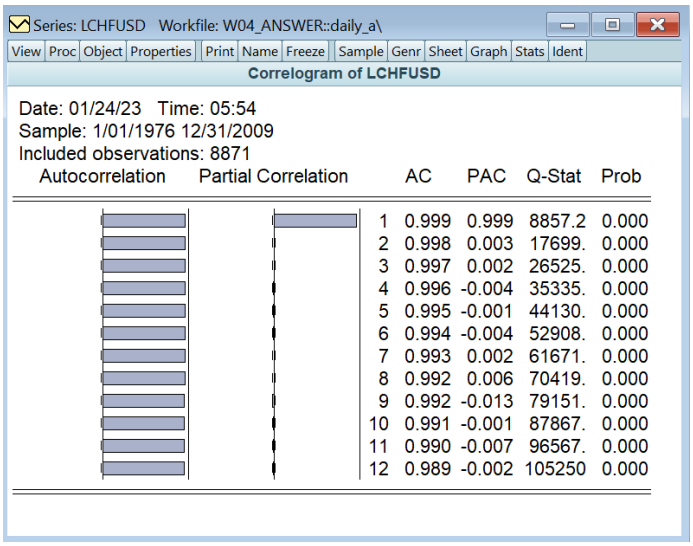
Question 2. Compute the autocorrelation functions (ACFs) and partial ACFs (PACFs) of the **lchfusd** series. Do the ACFs decline slowly or quickly as the lag lengths increase? Do the PACFs show a pronounced drop-off after a certain lag length?

Double click on the series **lchfusd**, choose "View→ Correlogram."

Or type in the command window:

Show lchfusd.correl(12)

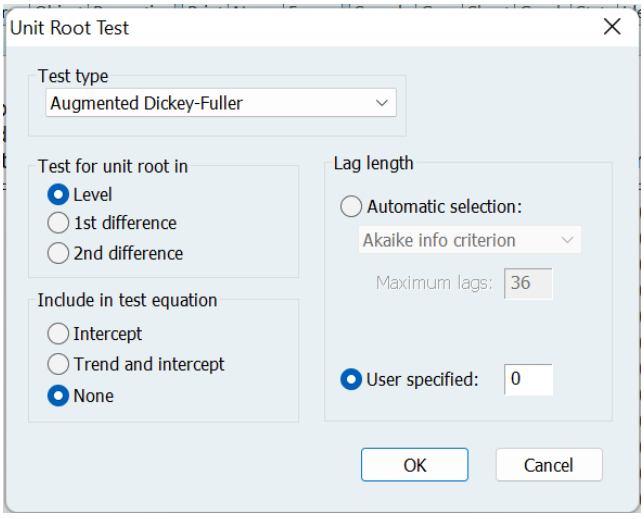
Then press enter.



Clearly, the autocorrelation function does not die out for this series, suggesting the presence of a unit root in the (log) level of the series.

Question 3. Perform *non-augmented* Dickey-Fuller unit root tests, for models without and with a constant term. What is the null and alternative hypotheses used in these tests? What are the p values of the test statistics? Do you reject the null for either of the two tests?

Click View→ Unit Root Test→Standard Unit Root Test.
(Lag length = 0 implies no lagged dependent variables are included as regressors; i.e. non-augmented DF case.)



Series: LOG(CHFUSD) Workfile: W03_DATA::daily_a\

View | Proc | Object | Properties | Print | Name | Freeze | Sample | Genr | Sheet | Graph | Stats | Ident

Augmented Dickey-Fuller Unit Root Test on LOG(CHFUSD)

Null Hypothesis: LOG(CHFUSD) has a unit root
Exogenous: None
Lag Length: 0 (Fixed)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-2.022329	0.0414
Test critical values:	1% level	-2.565223	
	5% level	-1.940860	
	10% level	-1.616676	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(LOG(CHFUSD))
Method: Least Squares
Date: 05/18/22 Time: 05:18
Sample: 1/01/1976 12/31/2009
Included observations: 8871

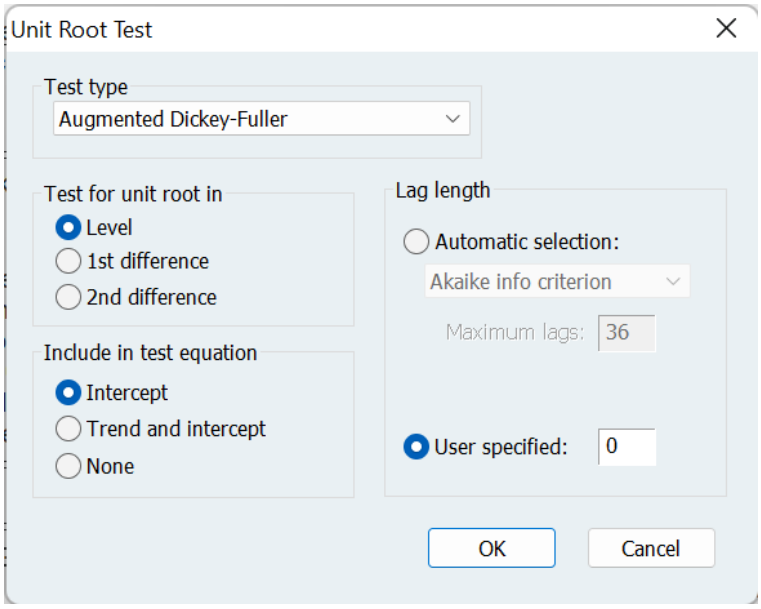
Variable	Coefficient	Std. Error	t-Statistic	Prob.
LOG(CHFUSD(-1))	-0.000313	0.000155	-2.022329	0.0432

R-squared	0.000257	Mean dependent var	-0.000105
Adjusted R-squared	0.000257	S.D. dependent var	0.007326
S.E. of regression	0.007325	Akaike info criterion	-6.994882
Sum squared resid	0.475950	Schwarz criterion	-6.994083
Log likelihood	31026.80	Hannan-Quinn criter.	-6.994610
Durbin-Watson stat	2.008431		

The null-hypothesis of the ADF tests is that the series has a unit root (i.e., the coefficient **lchfusd(-1)** is zero) in the above regression. The alternative is that the series is stationary around 0 (in the first case, when we don't have a constant) and around a constant mean (when we have a constant in the ADF regression).

The non-augmented ADF test (*without a constant*) suggests that **lchfusd** is stationary because the p-value of the test statistic is less than 0.05. However, this is likely to be a false rejection of the null because **lchfusd** has a positive mean, implying that it is a mistake to exclude the constant from the regression. Moreover, notice that the estimated coefficient on the lagged **lchfusd**, even though statistically significant, is rather *close to zero* – implying that the estimated coefficient on the lagged variable in the level representation is *near unity*.

Lastly, we cannot reject the null hypothesis of a unit root when we include a constant in the fitted ADF regression. But even this conclusion can be questioned because we do not allow for the possibility that the first difference of the **lchfusd** may be correlated when using the non-augmented Dickey-Fuller statistic (though for exchange rates, this correlation is typically rather weak. See below).



Series: LOG(CHFUSD) Workfile: W03_DATA::daily_a\

View | Proc | Object | Properties | Print | Name | Freeze | Sample | Genr | Sheet | Graph | Stats | Ident

Augmented Dickey-Fuller Unit Root Test on LOG(CHFUSD)

Null Hypothesis: LOG(CHFUSD) has a unit root
Exogenous: Constant
Lag Length: 0 (Fixed)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-1.788337	0.3868
Test critical values:	1% level	-3.430911	
	5% level	-2.861673	
	10% level	-2.566882	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(LOG(CHFUSD))
Method: Least Squares
Date: 05/18/22 Time: 05:20
Sample: 1/01/1976 12/31/2009
Included observations: 8871

Variable	Coefficient	Std. Error	t-Statistic	Prob.
LOG(CHFUSD(-1))	-0.000596	0.000333	-1.788337	0.0738
C	0.000161	0.000168	0.959998	0.3371

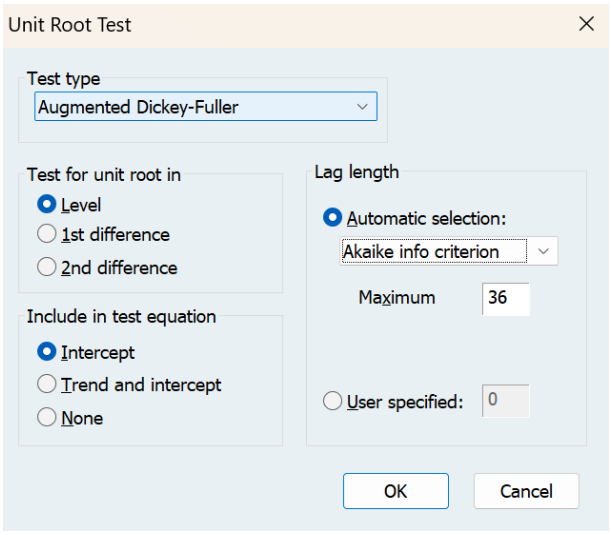
R-squared	0.000360	Mean dependent var	-0.000105
Adjusted R-squared	0.000248	S.D. dependent var	0.007326
S.E. of regression	0.007325	Akaike info criterion	-6.994761
Sum squared resid	0.475901	Schwarz criterion	-6.993162
Log likelihood	31027.26	Hannan-Quinn criter.	-6.994216
F-statistic	3.198149	Durbin-Watson stat	2.008070
Prob(F-statistic)	0.073756		

Question 4. Perform *augmented* DF tests on this series, again without and with a constant term in the model. Use the AIC (Akaike Information Criterion) to let EViews decide the number of lags, p, of Δyt that should be used in the ADF regression. Is the value of p chosen by the AIC

different from 0? Does your inference regarding the presence of a unit root depend on whether you use the ADF tests rather than the plain DF tests?

The following dialog box directs EViews to calculate 37 ADF regressions, with lags from 0 to 36 (inclusive). It selects (automatically) the ADF regression that has the smallest AIC, which in this case is 0.

Results are therefore the same as indicated previously with the unaugmented DF test.



Series: LOG(CHFUSD) Workfile: W03_DATA::daily_a\

View | Proc | Object | Properties | Print | Name | Freeze | Sample | Genr | Sheet | Graph | Stats | Ident

Augmented Dickey-Fuller Unit Root Test on LOG(CHFUSD)

Null Hypothesis: LOG(CHFUSD) has a unit root
Exogenous: None
Lag Length: 0 (Automatic - based on AIC, maxlag=36)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-2.022329	0.0414
Test critical values:	1% level	-2.565223	
	5% level	-1.940860	
	10% level	-1.616676	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(LOG(CHFUSD))
Method: Least Squares
Date: 05/18/22 Time: 05:26
Sample: 1/01/1976 12/31/2009
Included observations: 8871

Variable	Coefficient	Std. Error	t-Statistic	Prob.
LOG(CHFUSD(-1))	-0.000313	0.000155	-2.022329	0.0432

R-squared	0.000257	Mean dependent var	-0.000105
Adjusted R-squared	0.000257	S.D. dependent var	0.007326
S.E. of regression	0.007325	Akaike info criterion	-6.994882
Sum squared resid	0.475950	Schwarz criterion	-6.994083
Log likelihood	31026.80	Hannan-Quinn criter.	-6.994610
Durbin-Watson stat	2.008431		

Series: LOG(CHFUSD) Workfile: W03_DATA::daily_a\

View | Proc | Object | Properties | Print | Name | Freeze | Sample | Genr | Sheet | Graph | Stats | Ident

Augmented Dickey-Fuller Unit Root Test on LOG(CHFUSD)

Null Hypothesis: LOG(CHFUSD) has a unit root
Exogenous: Constant
Lag Length: 0 (Automatic - based on AIC, maxlag=36)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-1.788337	0.3868
Test critical values:	1% level	-3.430911	
	5% level	-2.861673	
	10% level	-2.566882	

*MacKinnon (1996) one-sided p-values.

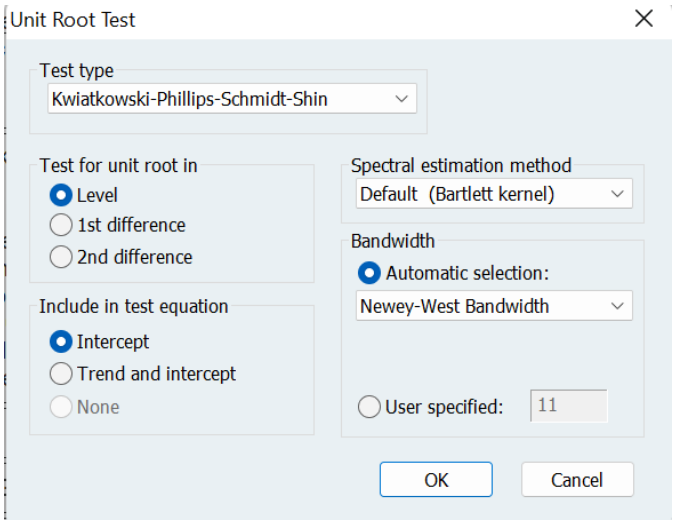
Augmented Dickey-Fuller Test Equation
Dependent Variable: D(LOG(CHFUSD))
Method: Least Squares
Date: 05/18/22 Time: 05:25
Sample: 1/01/1976 12/31/2009
Included observations: 8871

Variable	Coefficient	Std. Error	t-Statistic	Prob.
LOG(CHFUSD(-1))	-0.000596	0.000333	-1.788337	0.0738
C	0.000161	0.000168	0.959998	0.3371

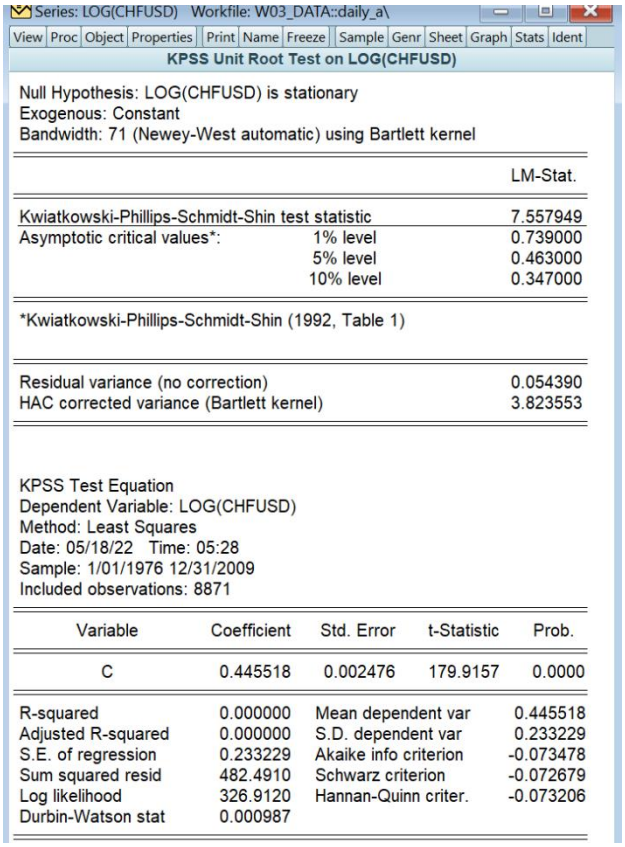
R-squared	0.000360	Mean dependent var	-0.000105
Adjusted R-squared	0.000248	S.D. dependent var	0.007326
S.E. of regression	0.007325	Akaike info criterion	-6.994761
Sum squared resid	0.475901	Schwarz criterion	-6.993162
Log likelihood	31027.26	Hannan-Quinn criter.	-6.994216
F-statistic	3.198149	Durbin-Watson stat	2.008070
Prob(F-statistic)	0.073756		

Question 5. Perform the KPPS test. The null hypothesis of stationarity is rejected: does this finding contradict the previous results?

Following the usual steps, we click on the "Test type" combo box and select KPSS:



Indeed, the null hypothesis is rejected, but recall that for the KPSS test the null is series stationarity so a rejection of the null hypothesis in this case confirms that there is a unit root in the series.



Series: LOG(CHFUSD) Workfile: W03_DATA::daily_a\

View | Proc | Object | Properties | Print | Name | Freeze | Sample | Genr | Sheet | Graph | Stats | Ident

KPSS Unit Root Test on LOG(CHFUSD)

Null Hypothesis: LOG(CHFUSD) is stationary
Exogenous: Constant
Bandwidth: 71 (Newey-West automatic) using Bartlett kernel

		LM-Stat.
Kwiatkowski-Phillips-Schmidt-Shin test statistic		7.557949
Asymptotic critical values*:	1% level	0.739000
	5% level	0.463000
	10% level	0.347000

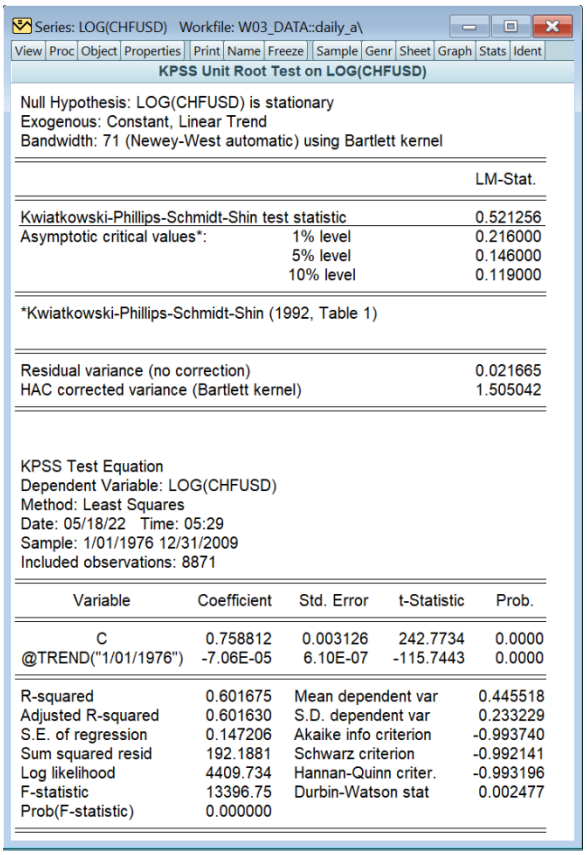
*Kwiatkowski-Phillips-Schmidt-Shin (1992, Table 1)

Residual variance (no correction)	0.054390
HAC corrected variance (Bartlett kernel)	3.823553

KPSS Test Equation
Dependent Variable: LOG(CHFUSD)
Method: Least Squares
Date: 05/18/22 Time: 05:28
Sample: 1/01/1976 12/31/2009
Included observations: 8871

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.445518	0.002476	179.9157	0.0000

R-squared	0.000000	Mean dependent var	0.445518
Adjusted R-squared	0.000000	S.D. dependent var	0.233229
S.E. of regression	0.233229	Akaike info criterion	-0.073478
Sum squared resid	482.4910	Schwarz criterion	-0.072679
Log likelihood	326.9120	Hannan-Quinn criter.	-0.073206
Durbin-Watson stat	0.000987		



Series: LOG(CHFUSD) Workfile: W03_DATA::daily_a\

View | Proc | Object | Properties | Print | Name | Freeze | Sample | Genr | Sheet | Graph | Stats | Ident

KPSS Unit Root Test on LOG(CHFUSD)

Null Hypothesis: LOG(CHFUSD) is stationary
Exogenous: Constant, Linear Trend
Bandwidth: 71 (Newey-West automatic) using Bartlett kernel

		LM-Stat.
Kwiatkowski-Phillips-Schmidt-Shin test statistic		0.521256
Asymptotic critical values*:	1% level	0.216000
	5% level	0.146000
	10% level	0.119000

*Kwiatkowski-Phillips-Schmidt-Shin (1992, Table 1)

Residual variance (no correction)	0.021665
HAC corrected variance (Bartlett kernel)	1.505042

KPSS Test Equation
Dependent Variable: LOG(CHFUSD)
Method: Least Squares
Date: 05/18/22 Time: 05:29
Sample: 1/01/1976 12/31/2009
Included observations: 8871

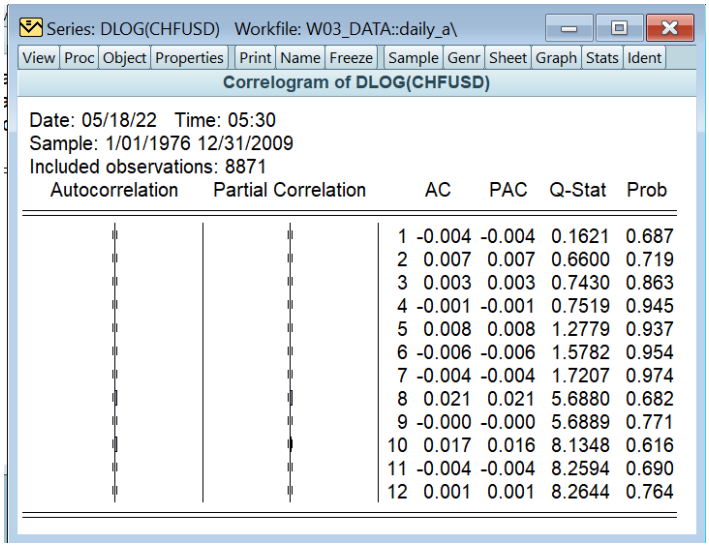
Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.758812	0.003126	242.7734	0.0000
@TREND("1/01/1976")	-7.06E-05	6.10E-07	-115.7443	0.0000

R-squared	0.601675	Mean dependent var	0.445518
Adjusted R-squared	0.601630	S.D. dependent var	0.233229
S.E. of regression	0.147206	Akaike info criterion	-0.993740
Sum squared resid	192.1881	Schwarz criterion	-0.992141
Log likelihood	4409.734	Hannan-Quinn criter.	-0.993196
F-statistic	13396.75	Durbin-Watson stat	0.002477
Prob(F-statistic)	0.000000		

Question 6. Forecasting with simple ARIMA models.

Part 1. Consider the autocorrelation functions (ACFs) and partial autocorrelation functions (PACFs) of d(lchfusd) since we found that lchfusd could be I(1).

This gives:



Thus, it does not seem to be necessary to include additional lags for the dependent variables.

Part 2. Estimate the model dlog(chfusd)=c for the period 1/1/1976 and 11/30/2009.

The required EViews commands are:

smpl 1976M1 2009M11
equation eq01_constant.ls dlog(chfusd) c

Equation: EQ01_CONSTANT Workfile: W04_ANSWER::daily_a\

View | Proc | Object | Print | Name | Freeze | Estimate | Forecast | Stats | Resids

Dependent Variable: DLOG(CHFUSD)
Method: Least Squares
Date: 01/24/23 Time: 06:46
Sample: 1/01/1976 11/30/2009
Included observations: 8848

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	-0.000108	7.79E-05	-1.389914	0.1646

R-squared	0.000000	Mean dependent var	-0.000108
Adjusted R-squared	0.000000	S.D. dependent var	0.007330
S.E. of regression	0.007330	Akaike info criterion	-6.993533
Sum squared resid	0.475357	Schwarz criterion	-6.992732
Log likelihood	30940.39	Hannan-Quinn criter.	-6.993261
Durbin-Watson stat	2.008145		

Notice that by fitting an ARMA model to the first log-difference of **chfusd** we are effectively assuming the existence of a unit root in the level of the exchange rate data.

The model is consistent with the random walk model, as can be seen by estimating:
equation eq02_random_walk.ls log(chfusd) c log(chfusd(-1))

Equation: EQ02_RANDOM_WALK Workfile: W04_ANSWER::daily_a\

View | Proc | Object | Print | Name | Freeze | Estimate | Forecast | Stats | Resids

Dependent Variable: LOG(CHFUSD)
Method: Least Squares
Date: 01/24/23 Time: 06:41
Sample: 1/01/1976 11/30/2009
Included observations: 8848

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.000147	0.000169	0.873323	0.3825
LOG(CHFUSD(-1))	0.999428	0.000335	2983.071	0.0000

R-squared	0.999007	Mean dependent var	0.446598
Adjusted R-squared	0.999007	S.D. dependent var	0.232566
S.E. of regression	0.007329	Akaike info criterion	-6.993637
Sum squared resid	0.475200	Schwarz criterion	-6.992035
Log likelihood	30941.85	Hannan-Quinn criter.	-6.993092
F-statistic	8898715.	Durbin-Watson stat	2.007658
Prob(F-statistic)	0.000000		

In addition, all tests suggest that residuals from the differenced model (*eq01_constant*) do not suffer from serial correlation (up to 12 lags), confirming that the estimated model is appropriate.

Equation: UNTITLED Workfile: W03_DATA::daily_a\

View | Proc | Object | Print | Name | Freeze | Estimate | Forecast | Stats | Resids

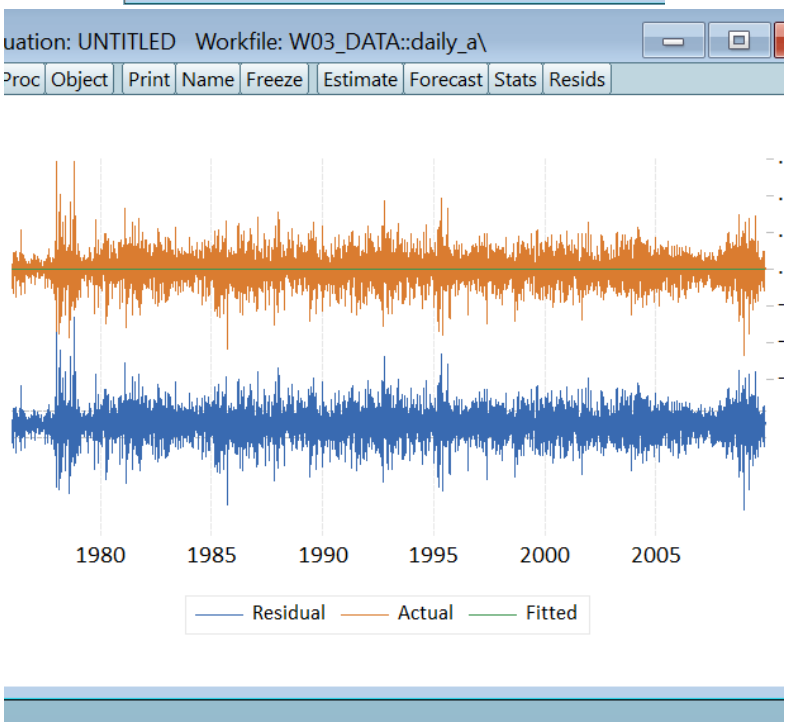
Breusch-Godfrey Serial Correlation LM Test:
Null hypothesis: No serial correlation at up to 12 lags

F-statistic	0.689541	Prob. F(12,8835)	0.7633
Obs*R-squared	8.278908	Prob. Chi-Square(12)	0.7630

Test Equation:
Dependent Variable: RESID
Method: Least Squares
Date: 05/18/22 Time: 05:39
Sample: 1/01/1976 11/30/2009
Included observations: 8848
Presample missing value lagged residuals set to zero.

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	4.60E-09	7.79E-05	5.90E-05	1.0000
RESID(-1)	-0.003875	0.010639	-0.364232	0.7157
RESID(-2)	0.006677	0.010639	0.627556	0.5303
RESID(-3)	0.003232	0.010638	0.303855	0.7612
RESID(-4)	-0.001029	0.010640	-0.096720	0.9230
RESID(-5)	0.007138	0.010638	0.671035	0.5022
RESID(-6)	-0.006037	0.010638	-0.567438	0.5704
RESID(-7)	-0.004222	0.010639	-0.396832	0.6915
RESID(-8)	0.021449	0.010639	2.016165	0.0438
RESID(-9)	4.71E-05	0.010641	0.004426	0.9965
RESID(-10)	0.016481	0.010642	1.548629	0.1215
RESID(-11)	-0.003952	0.010644	-0.371302	0.7104
RESID(-12)	0.000929	0.010644	0.087293	0.9304

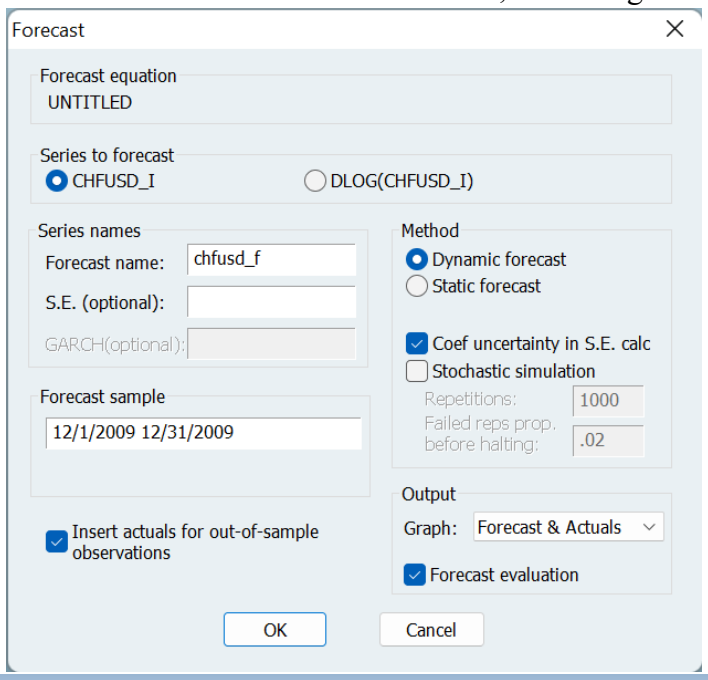
R-squared	0.000936	Mean dependent var	1.43E-18
Adjusted R-squared	-0.000421	S.D. dependent var	0.007330
S.E. of regression	0.007332	Akaike info criterion	-6.991757
Sum squared resid	0.474912	Schwarz criterion	-6.981343
Log likelihood	30944.53	Hannan-Quinn criter.	-6.988210
F-statistic	0.689541	Durbin-Watson stat	1.999979
Prob(F-statistic)	0.763264		

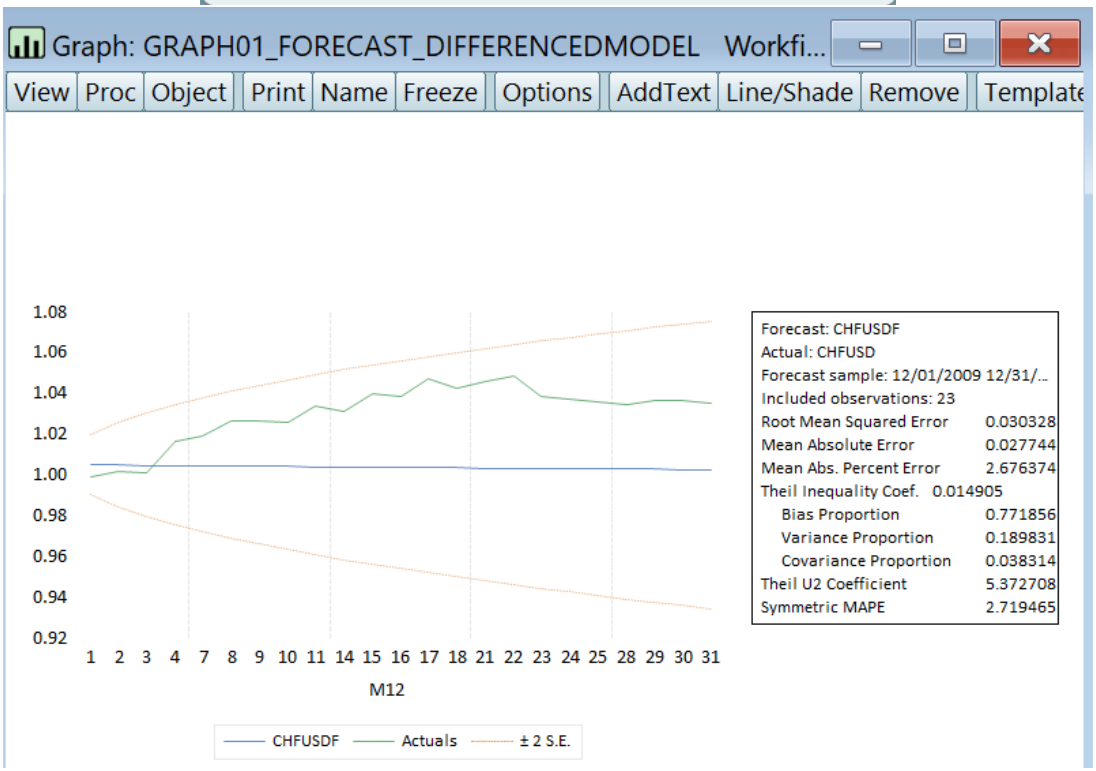


Now invoke Proc→Automatic ARIMA forecasting!

Part 3. Forecast **chfusd** for each day in December 2009. For this, use the sample 12/1/2009 - 12/31/2009. Plot the forecast series and the actual series over the forecast period

Notice that the target of the forecast is the level of the series, not its log level.





Not a very exciting outcome! But this result is to be expected since exchange rates are notoriously difficult to predict.

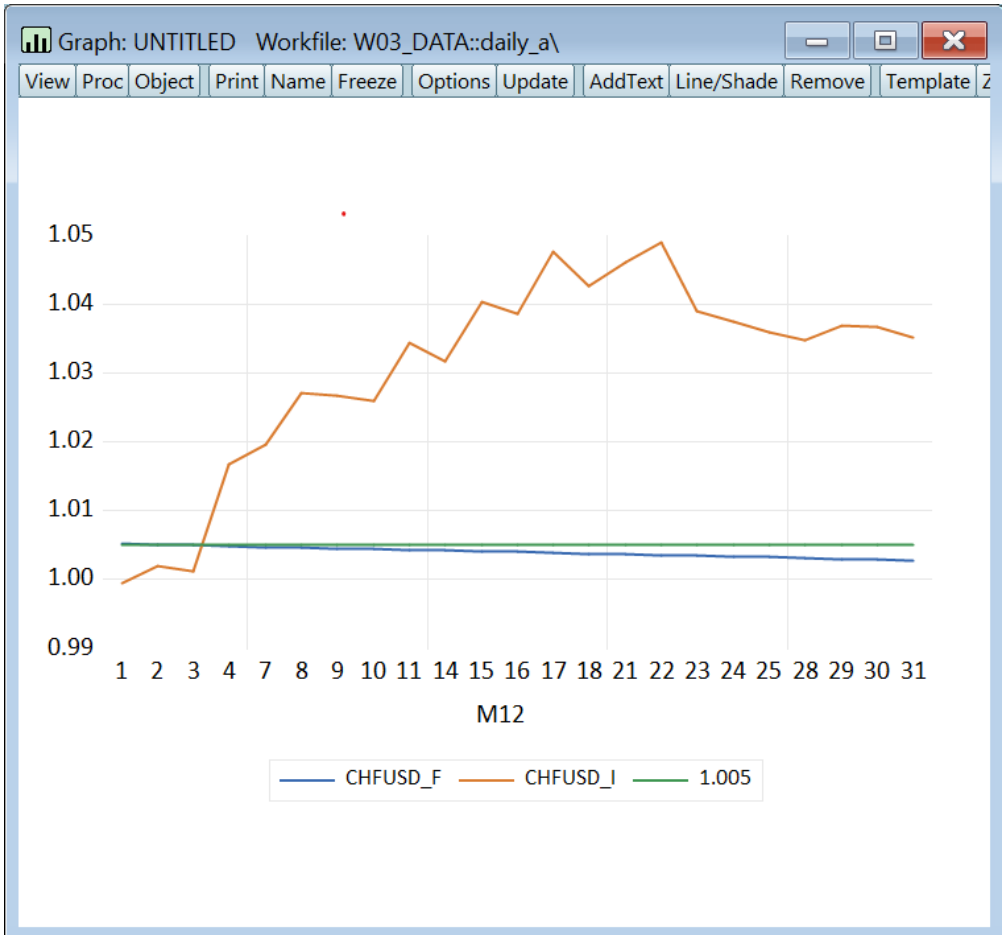
Part 4. Now let's compare it with the pure-random walk model, which assumes zero change from the end of November 2009 onwards (i.e., the optimal forecast for December 31, 2009 is

actually the value of the exchange rate on November 30, 2009). The value of the exchange rate on November 30, 2009 was 1.005.

smpl 12/1/2009 12/31/2009
plot chfusd chfusdf 1.005

It's rather clear that the no-change pure random walk model would yield a superior forecast assessment compared to the ARMA model for the first difference of the exchange rate.

This result is fairly common for exchange rate data, reflecting the common finding that nominal exchange rates are mostly white-noise, unpredictable series.



Takeaways from the Workshop

1. **Stationarity in Time Series:**
   - Recognizing stationarity is crucial for effective time series modeling. Non-stationary series often exhibit stochastic trends, requiring transformation (e.g., differencing) to achieve stationarity.
   - Tests like the Augmented Dickey-Fuller (ADF) and KPSS are essential for diagnosing stationarity, with careful attention to their null hypotheses: ADF tests for the presence of a unit root (non-stationarity), while KPSS tests for stationarity.
2. **Unit Root Tests and Their Interpretation:**
   - The distinction between rejecting or failing to reject null hypotheses in the ADF and KPSS tests was emphasized. A high KPSS test statistic leading to rejection of stationarity aligns with ADF findings (in case a constant is included) when a unit root is indicated.
3. **Modeling Exchange Rates:**
   - Exchange rate series like LCHFUSD demonstrated non-stationary behavior in levels but stationarity in first differences (I(1)).
   - ARIMA modeling, including model selection through ACF and PACF analysis, revealed challenges in forecasting exchange rates due to their unpredictable nature, aligning with findings of random walk behavior.
4. **Practical Challenges in Model Estimation**:
   - The importance of including relevant components (e.g., constant terms) in ADF tests to avoid misleading results was highlighted.
   - Forecasting results illustrated that while ARIMA models can capture short-term dynamics, simpler benchmarks like the random walk often outperform in forecasting nominal exchange rates.
5. **Conclusion**
   - This session underscored the critical role of correctly diagnosing stationarity and appropriately modeling time series. Exchange rate forecasting remains inherently challenging, with simpler models like the random walk often outperforming more sophisticated ARIMA approaches.